

# CSE221: Algorithms

## Lab-03

### Spring 2026

#### Assignment-03 Links:

##### codeforces Invitation Link:

<https://codeforces.com/contestInvitation/4a44ecb15e94edc9b764e2f8275344030b17829a>

##### Alternative Link:

<https://codeforces.com/contests/673915>

[The alternative link will **only work** after you have successfully registered using the first link.]

**Deadline:** Mar 8, 2026 2:00 PM

After the deadline ends, you can submit your solution by entering the contest as a Virtual Participant for practice purposes. **However, please note, submissions after the deadline are not accepted for grading.**

#### Learning Outcome:

In this lab, we will explore the **Divide and Conquer** paradigm, which is a fundamental algorithmic technique for solving problems efficiently by breaking them into smaller subproblems. Through hands-on problem-solving, students will develop a deeper understanding of recursion, algorithm design, and optimization.

#### Understanding the Divide and Conquer Approach

- Learn how to break a problem into **smaller independent subproblems**, solve them recursively, and combine the results.
- Understand when and why **Divide and Conquer** is preferable over brute-force approaches.
- Apply recurrence relations and the Master Theorem to analyze **time complexity**.